BMP3 Sensor Driver

API Reference

File: **bmp3.h** | Version: **v2.0.6** | Date: **2022-04-01**

Overview

The **BMP3** driver provides a portable C API for the Bosch BMP390 and BMP388 barometric pressure and temperature sensors. It handles sensor initialisation, raw data retrieval over I2C or SPI, and compensation of raw ADC values into physical units (Pa and °C) using the factory-calibrated trimming parameters stored in the device NVM.

Key features:

- **Bus-agnostic** – Works over I2C and SPI via user-supplied read/write callbacks.
- **Self-contained** – Single header, no dynamic allocation, C99-compatible.
- **Compensation** – Full Bosch double-precision compensation formula included.
- **Error reporting** – Signed 8-bit return codes for easy chained error checking.

Return Codes

All API functions return `int8_t`. A return value of **BMP3_OK (0)** indicates success; negative values indicate errors.

Macro	Value	Description
`BMP3_OK`	`0`	Operation completed successfully.
`BMP3_E_NULL_PTR`	`-1`	A required pointer argument was NULL.
`BMP3_E_COMM_FAIL`	`-2`	Bus communication (I2C/SPI) failed.
`BMP3_E_INVALID_LEN`	`-3`	An invalid data length was supplied.
`BMP3_E_DEV_NOT_FOUND`	`-4`	No BMP3 device detected on the bus.

Data Structures

struct bmp3_calib_data

Calibration data structure for the BMP3 sensor.

> Holds the trimming parameters read from the sensor's NVM. These are used internally for compensating raw ADC readings.

Fields:

Type / Name	Description
`double par_t1`	Calibration coefficient T1
`double par_t2`	Calibration coefficient T2
`double par_t3`	Calibration coefficient T3
`double par_p1`	Calibration coefficient P1
`double par_p2`	Calibration coefficient P2
`double t_lin`	Compensated temperature (linear intermediate)

struct bmp3_uncomp_data

Uncompensated (raw) ADC sensor data.

> Populated by bmp3_get_uncomp_data() and passed to bmp3_compensate_data().

Fields:

Type / Name	Description
`uint32_t pressure`	Raw pressure ADC value
`uint32_t temperature`	Raw temperature ADC value

struct bmp3_data

Compensated sensor data in SI units.

> Populated by bmp3_compensate_data() for application use.

Fields:

Type / Name	Description
`double pressure`	Compensated pressure in Pascals (Pa)
`double temperature`	Compensated temperature in degrees Celsius (°C)

Function Reference

Initialise the BMP3 sensor.

```
int8_t bmp3_init(void *dev)
```

Resets the device, reads the chip-ID to confirm the sensor is present, then loads calibration coefficients from NVM into the device structure.

Parameters:

Parameter	Description
`[in,out] dev`	Pointer to the initialised device structure.

Returns:

BMP3_OK on success; BMP3_E_NULL_PTR, BMP3_E_COMM_FAIL, or BMP3_E_DEV_NOT_FOUND on failure.

Read raw pressure and temperature from the sensor.

```
int8_t bmp3_get_uncomp_data(struct bmp3_uncomp_data *uncomp_data, const void *dev)
```

Issues a forced measurement and retrieves the 24-bit ADC results over the configured bus (I2C or SPI).

Parameters:

Parameter	Description
`[out] uncomp_data`	Pointer to store the raw ADC values.
`[in]  dev`	Pointer to the device structure.

Returns:

BMP3_OK on success; negative error code on failure.

Compensate raw sensor data.

```
int8_t bmp3_compensate_data(struct bmp3_data *comp_data, const struct bmp3_uncomp_data *uncomp_dat
```

Applies the Bosch double-precision compensation formula to convert raw ADC values to physical units (Pa and °C).

Parameters:

Parameter	Description
`[out] comp_data`	Pointer to store compensated output (Pa / °C).
`[in]  uncomp_data`	Raw ADC values from bmp3_get_uncomp_data().
`[in]  calib_data`	Calibration coefficients loaded during bmp3_init().

Returns:

BMP3_OK on success; BMP3_E_NULL_PTR if any pointer is NULL.

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